Customer journey and risk comparison

JourneyGraph AI | C003 and DEMO-IMPORT-001 | 5 November 2026

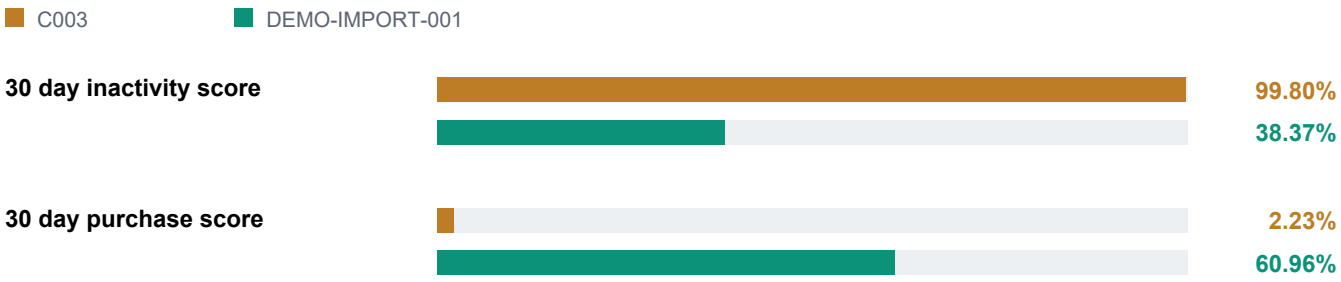
C003 needs a re-engagement review; DEMO-IMPORT-001 merits a relevant product follow-up. Both customers have four recorded interactions, but C003 stopped after browsing and has no purchase. The imported customer recently completed a $149 purchase. This case study demonstrates how temporal evidence supports different model scores and actions.

All records are synthetic demonstration data. Prediction percentages are uncalibrated model scores, not verified real-world probabilities.

Comparison	C003	DEMO-IMPORT-001
Segment	Inactive	New buyers
Recorded interactions	4	4
Channels and transitions	Social; 0 transitions	Website and mobile; 1 transition
Purchase history	0 purchases; $0	1 purchase; $149
Days since latest activity	161 completed days	3 completed days
Inactivity score and band	99.80% | High	38.37% | Low
Purchase score and band	2.23% | Low	60.96% | Medium
Recommended review	Re-engagement opportunity	Relevant product follow-up
Priority	High	Normal

Recency is measured against the latest dataset event, 4 November 2026 at 19:20 UTC, rather than the report creation time.

Prediction score comparison



Presentation takeaway: total activity counts can look similar while event sequence, recency, and purchase evidence point to very different engagement states.

Recorded customer journeys

The following timelines use UTC event time. They reconstruct observed behavior; they do not establish the customer's intention or the cause of disengagement.

C003 browsing without a recorded purchase

Event time UTC	Interaction	Product	Channel
24 May 2026 11:43	Website visit	P005 Desk lamp	social
24 May 2026 11:47	Product view	P005 Desk lamp	social
26 May 2026 20:11	Website visit	P004 Everyday sneakers	social
26 May 2026 20:15	Product view	P004 Everyday sneakers	social

Evidence IDs EVT000107 to EVT000110. The two browsing visits span 2.36 days. There are no cart or purchase events, and no events in the final 30-day feature window.

Interpretation. C003 disengaged after browsing. Calling this cart abandonment would be unsupported because no cart was recorded. The event channel is social even though the interaction type is website_visit; the report preserves the stored fields.

DEMO IMPORT 001 a completed purchase journey

Event time UTC	Interaction	Product	Channel
1 Nov 2026 10:00	Website visit	P001 Studio headphones	website
1 Nov 2026 10:04	Product view	P001 Studio headphones	website
1 Nov 2026 10:08	Add to cart	P001 Studio headphones	mobile
1 Nov 2026 10:12	Purchase $149	P001 Studio headphones	mobile

Evidence IDs DEMO-IMPORT-E01 to DEMO-IMPORT-E04. The sequence lasts 12 minutes and contains one website-to-mobile transition.

Interpretation. The recorded cart was followed by a purchase. P001 appears in both the product view and purchase evidence, making it a relevant subject for a human-reviewed follow-up.

Temporal graph verification

For DEMO-IMPORT-001, the browser showed 5 entities and 8 relationships: one entity fact and one channel fact for each event. A valid-at query before the journey returned zero facts. A known-at query before import also returned zero facts, demonstrating the separate event-time and transaction-time filters.

Each purchase fact retains the source event ID, valid-from timestamp, server-recorded timestamp, channel, and synthetic-demo provenance. An open-ended event fact remains in the cumulative graph; it does not mean the activity continues indefinitely.

Predictions and recommended actions

Both customers were scored by the same model family and evaluated population history. Inactivity means no tracked events in the next 30 days. Conversion means a purchase in the next 30 days, including a possible repeat purchase for the imported customer.

Why the scores differ

C003. About 162 days of recency and zero events, purchases, carts, support events, or active channels in the final 30-day feature window. Recency is the largest positive contribution to the inactivity model and the largest negative contribution to the purchase model.

DEMO-IMPORT-001. About 3.38 days of recency, four recent events, one purchase, one cart event, and two active channels. This produces a lower inactivity score and a higher purchase score. Feature contributions explain the fitted model, not causal effects.

Evaluated target	ROC AUC	Balanced accuracy	Brier score
30 day inactivity	0.8550	77.60%	0.1557
30 day purchase	0.7796	75.56%	0.1813

Evaluation: 120 training customers and 120 holdout customers. Training features end 5 September 2026; their 30-day labels end 5 October. Holdout features end 5 October; outcomes end 4 November. The exact cutoffs are 19:20 UTC. Both models use StandardScaler and balanced logistic regression.

ROC AUC measures ranking separation; balanced accuracy averages sensitivity across the two classes; lower Brier scores indicate smaller squared probability errors. These metrics assess a single synthetic temporal holdout, not production validity.

C003 recommendation

Review re-engagement opportunity | High priority. The rule triggers because recency exceeds 30 days. The proposal cites EVT000107 to EVT000110 and the observed products P005 and P004. The 99.80% inactivity score supports the risk comparison, but it does not directly choose this action.

DEMO IMPORT 001 recommendation

Review relevant product follow-up | Normal priority. The customer is recently active and the latest cart has a subsequent purchase. The follow-up rule therefore applies instead of inactivity re-engagement or cart assistance. The proposal cites DEMO-IMPORT-E01 to DEMO-IMPORT-E04 and product P001.

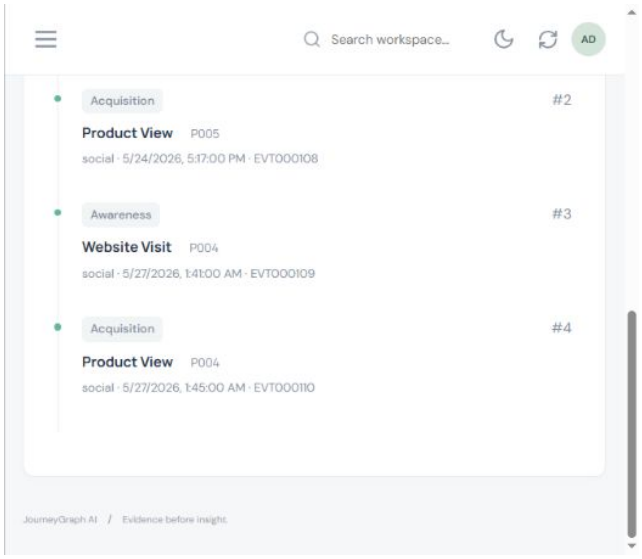
Human review and limits

Both actions remain pending_human_review. No message or intervention was sent, and expected uplift was not estimated. Scores are uncalibrated and have no uncertainty intervals. Customers can appear in both time windows; missing tracking can resemble churn. C003 never purchased, so high inactivity should not be described as confirmed loss of a paying customer.

Browser evidence and execution trace

Matching screenshot excerpts show the final recorded actions for each customer. The browser displays local India time; the chronology tables on page 2 use UTC. Screenshots preserve the original application display.

C003 final browsing events	DEMO-IMPORT-001 final purchase events



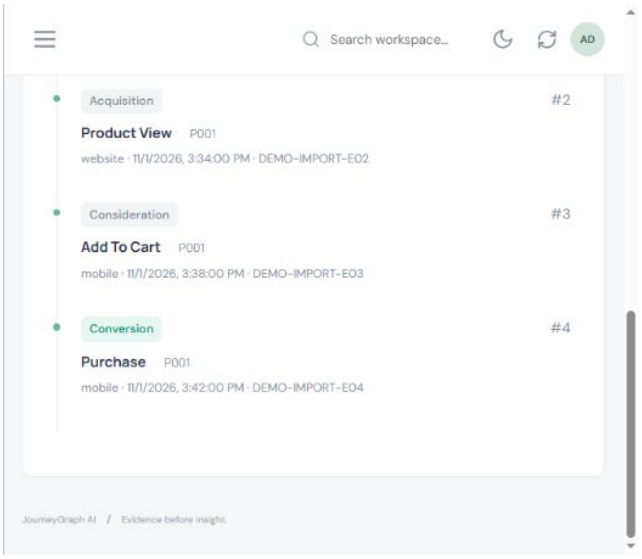


Figure 1 C003 ends at a product view without a recorded cart or purchase. Figure 2 DEMO-IMPORT-001 ends with add to cart and purchase. The full four-event timelines are reproduced on page 2.

Verified agent executions

All nine agents completed in both runs: profiling, tracking, temporal graph, behavior, segmentation, prediction, recommendation, insight generation, and orchestration. Insight generation used the labeled local evidence summary; no external LLM was called.

Customer	Status	Completed agents	Execution time IST
C003	Completed	9 of 9	09:54:07 to 09:54:10
DEMO-IMPORT-001	Completed	9 of 9	09:48:26 to 09:48:29

C003 execution 25b23238-ba93-4111-bef1-0a1ff1ce2cc8
DEMO-IMPORT-001 execution b11ffaa0-c556-4bdc-b6c2-2a310581fa18

Source evidence: docs/browser-workflow/high-risk-C003.json and agent-results.json; linked sample event files; product catalog in data/samples/products.csv. Screenshots captured from the running local application on 5 November 2026.